Maximiliano Zorondo

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Langs: Spanish, English, Italian
c++, bash, LaTeX, python

I am an eclectic researcher, I have worked in complex systems, plasma physics, optics, educational research, and now **economics** as well as in simulations, data analysis, theory, and experiments. As I feel profound political responsibility, I have moved away from physics, and currently my primary interests are researching the **Unconditional Basic Income** (UBI) and promoting it.

Education

2023/2024–Ongoing **Innovation and Management, Ph.D.**,
Università Cattolica del Sacro Cuore.
- Working on a literature review on the potential of a UBI as a driver for innovation
- Studing natural experiments of policies similar to a UBI during COVID-19
- Modeling the introduction and long-term effects of a UBI using Agent-based Models

Mar 2021 **Physics, M.Sc.**,
Universidad de Chile.
- Developed a statistical mechanical model of the scattering of electromagnetic waves (Thomson Scattering) from a nonequilibrium plasma
- Designed and implemented an experimental setup for Thomson Scattering in a Plasma Focus discharge of hundreds of joules

Dec 2015 **Physics, B.Sc.**,
Universidad de Chile.
- Studied the solar activity through sunspot evolution using complex networks

Experience

2015–2023 **Research Assistant**,
Research Center on the Intersection in Plasma Physics, Matter and Complexity (P2mc), Chilean Nuclear Energy Commission.
- Characterized for the first time the complete fast dynamics of Multipurpose Generator in Plasma Focus configuration on a regime of high total neutron output through shadowgraphy
- Implemented simultaneous Mach-Zendher interferometer and shadowgraphy in Multipurpose Generator Plasma Focus device
- Characterized neutron emission of Multipurpose Generator Plasma Focus device

Mar 2021–Sep 2021 **Research Assistant**,
Faculty of Science, University of Chile.
- Constructed and validated content of multimodal analytical rubric for assessing the quality of a laboratory physics report

Jan 2015 **Summer Intern**,
Optics and Plasma Physics Group, Institute of Physics, Pontifical Catholic University of Chile.

Aug 2014–Dec 2014 **Intern**,
Non-linear Optics Group, Faculty of Science, University of Chile.

Mar 2014–Dec 2014 **Intern**,
Group of Complex Systems, Faculty of Science, University of Chile.

Publications

- Pavez, C., Zorondo, M., Pedreros, J., Sepúlveda, A., Soto, L., Avaria, G., Bora, B., & Jain, J. (2022). New evidence about the nature of plasma filaments in plasma accelerators of type plasma-focus. *Plasma Physics and Controlled Fusion*, **65**(1), 015003.
- Zorondo, M., Pavez, C., & Muñoz, V. (2022). Model of Thomson scattering from z-pinch plasma: Application in

experimental design for Plasma Focus. *Results in Physics*, **40**, 105831.

- Bruzzone, H., Acuña, H., Barbaglia, M., Clausse, A., Milanese, M., Pavez, C., Avaria, G., Pedreros, J., Sepúlveda, A., Rojas, C., Zorondo, M., & Soto, L. (2018). Physical reasoning to synchronize electrical signals and related diagnostics in plasma focus devices. *Journal of Fusion Energy*, **37**, 45-50.

Teaching Experience

Lecturer

Spr 2022 **Advanced Experimental Methods**, *Faculty of Science, Metropolitan University of Educational Sciences*.

- As a result of the course, two groups of students under my guidance presented posters at the XXIII Simposio de la Sociedad Chilena de Física conference

Teaching Associate

Spr 2016 **Mathematical Methods for Physics II**, *Faculty of Sciences, University of Chile*.

Aut 2016 **Advanced Programing and Numerical Methods**, *Faculty of Sciences, University of Chile*.

Aut 2015 **Advanced Programing and Numerical Methods**, *Faculty of Sciences, University of Chile*.

Spr 2014 **Programing and Numerical Methods**, *Faculty of Sciences, University of Chile*.

Spr 2013 **Programing and Numerical Methods**, *Faculty of Sciences, University of Chile*.

Teaching Assistant

Spr 2020 **Experimental Methods for Electromagnetism**, *Vice presidency of Academic Affairs, University of Chile*.

Aut 2020 **Experimental Methods for Mechanics**, *Vice presidency of Academic Affairs, University of Chile*.

Spr 2018 **Experimental Methods for Mechanics**, *Faculty of Sciences, University of Chile*.

Aut 2018 **Experimental Methods for Optics**, *Faculty of Sciences, University of Chile*.

Spr 2017 **Experimental Methods for Mechanics**, *Faculty of Sciences, University of Chile*.

Spr 2017 **Experimental Methods for Mechanics**, *Vice presidency of Academic Affairs, University of Chile*.

Aut 2017 **Experimental Methods for Mechanics II**, *Faculty of Sciences, University of Chile*.

Aut 2016 **Calculus of Several Variables**, *Department of Mathematics and Engineering Sciences, Bernardo O'Higgins University*.

Aut 2016 **Calculus II**, *Department of Mathematics and Engineering Sciences, Bernardo O'Higgins University*.

Aut 2016 **Algebra I**, *Department of Mathematics and Engineering Sciences, Bernardo O'Higgins University*.

Conferences

Oral

Dec 2021 **Thomson Scattering Model for Z-pinch Plasma: Experimental Design for Implementation in Plasma Focus**, M. Zorondo, C. Pavez, and V. Muñoz. *14th International Conference on Plasma Science and Applications*, Online.

Dec 2021 **Dynamic characteristics of a low-energy plasma focus discharge on a regime of high total neutron output**, C. Pavez, M. Zorondo, A. Sepúlveda, J. Pedreros, G. Avaria, J. Moreno, B. Bora, S. Davis, J. Jain, and L. Soto. *14th International Conference on Plasma Science and Applications*, Online.

Dec 2021 **Low energy Plasma focus discharges in a high efficient neutron production regime**, C. Pavez, J. Pedreros, M. Zorondo, A. Sepúlveda, L. Orellana, J. Jain, J. Moreno, S. Davis, G. Avaria, B. Bora, L. Soto, F. Molina, B. Parra, J. Romero-Barrintos, and A. Tarifeño-Saldivia. *IEEE Pulsed Power Conference & Symposium of Fusion Engineering*, Online.

Nov 2020 **Modelo de dispersión de radiación de Thomson para Plasma Focus [*Thomson Scattering Model for Plasma Focus*]**, M. Zorondo, C. Pavez, and V. Muñoz. *XXII Simposio de la Sociedad Chilena de Física*, Online.

Nov 2020 **Efectos refractivos de objetos de axisimétricos: Estudio comparativo usando teoría de rayos y teoría de difracción [*Refractive effects of axisymmetric objects: Comparative study using ray and diffraction theories*]**, J. Pedreros, C. Pavez, M. Zorondo, G. Avaria, J. Moreno, B. Bora, S. Davis, J. Jain, and L. Soto. *XXII Simposio de la Sociedad Chilena de Física*, Online.

Nov 2020 **Neutron spectroscopy of a pulsed (d,d) reaction in the Multipurpose Plasma Generator device at CCHEN**, F. Molina, B. Parra, A. Tarifeño-Saldivia, C. Pavez, P. Aguilera, L. Orellana, M. Zorondo, G. Avaria, F. López-Usquiano, J. Romero-Barrientos, A. Ruiz, M. Zambra, B. Bora, S. Davis, J. Jain, J. Moreno, L. Soto, and H. F. Arellano. *XXII Simposio de la Sociedad Chilena de Física*, Online.

Nov 2020 **Caracterización de la dinámica de una lámina de corriente en un acelerador coaxial de plasma: Simulaciones y experimentos [*Dynamic characterization of a current sheet in a coaxial plasma accelerator: Simulation and experiments*]**, C. Pavez, A. Sepúlveda, J. Pedreros, M. Zorondo, G. Avaria, J. Moreno, B. Bora, S. Davis, J. Jain, L. Soto, and A. Clausse. *XXII Simposio de la Sociedad Chilena de Física*, Online.

Nov 2018 **Filamentaciones de Plasma en Aceleradores de Plasma Tipo Plasma Foco: Origen y Evolución [*Plasma filaments in plasma accelerators of the Plasma Focus type: Origin and Evolution*]**, C. Pavez, A. Sepúlveda, J. Pedreros, M. Zorondo, G. Avaria, J. Moreno, B. Bora, S. Davis, J. Jain, and L. Soto. *XXI Simposio de la Sociedad Chilena de Física*, Antofagasta, Chile.

Dec 2016 **Characterization of a plasma focus discharge under different regimes of input power density**, A. Sepúlveda, C. Rojas, M. Zorondo, J. Pedreros, C. Pavez C, H. Bruzzone, J. Moreno, B. Bora, S. Davis, M. Inestrosa-Izurieta, and L. Soto. *XX Simposio de la Sociedad Chilena de Física*, Santiago, Chile.

Dec 2016 **Three-frame Digital Interferometry Using a Single Digital Record System**, J. Pedreros, M. Zorondo, C. Pavez, G. Avaria, and L. Soto. *XX Simposio de la Sociedad Chilena de Física*, Santiago, Chile.

Poster

Nov 2023 **Mirando el mundo a través de un pequeño agujero: Experimento ilustrativo para educación media de formación de imágenes a través de una cámara oscura [*Looking at the world through a pinhole: Camera obscura illustrative experiment for high school students*]**, S. Carrasco Basso, F. Pardo Aránguiz, M. Zorondo, C. Pavez, and D. López. *XXIII Simposio de la Sociedad Chilena de Física*, Valparaiso, Chile.

Nov 2023 **Experimento ilustrativo para educación media de las características refractivas de elementos translúcidos [*Illustrative experiment for high school students on the refractive characteristics of translucent materials*]**, A. López Monsalve, D. Valenzuela Abarca, M. Zorondo, and C. Pavez. *XXIII Simposio de la Sociedad Chilena de Física*, Valparaiso, Chile.

Nov 2018 **Dinámica y emisión de una descarga Plasma Focus: Escalamiento en rango extendido de densidad de potencia de entrada [*Dynamic and Emission of a Plasma Focus Discharge: Scaling on Extended Range of Power Input*]**, M. Zorondo, C. Pavez, A. Sepúlveda, J. Pedreros, G. Avaria, J. Moreno and L. Soto. *XXI Simposio de la Sociedad Chilena de Física*, Antofagasta, Chile.

Nov 2014 **Estudio de actividad solar usando redes complejas [*Study of Solar Activity Using Complex Networks*]**, M. Zorondo and V. Muñoz. *XIX Simposio de la Sociedad Chilena de Física*, Concepción, Chile.